CO2 - AT3 - Case Study

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Reg. No: 192424420 Data Handling & Visualization

Case-Study: 1

a) **Nature of Dataset:** 2 Continous variables (fertilizer/yield) → bivariate. Matters because it rules out univariate / geospatial charts, points to Scatter (line) regression.

b) **Scatter:** Shows true relationship; exposes outlier. (obs 4 : 5.2 → 5.1)
↳ **Weak:** no sequence, sparse with 8 points.

Line: easy to read if ordered; visually connects trends.
↳ **Weak:** No Natural sequence here (fertilizer ≠ time) implies false continuity / trend

i) High fertilizer (14.2 - 14.9) → high yield (4.6 - 4.7), general positive correlation

Outlier: obs 4 (5.2 fert, 2.1 yield) lower than obs 3 (1.7 fert, 2.6 yield) - break trends

d) Recommendation: Scatter plot

→ built for x-y correlation, not sequence.

→ isolates outlier obs 4 clearly.

→ avoids false interpolation between unrelated fertilizer levels.

e) Limitation & fix

→ 8 points is too few to confirm statistical correlation strength (no trendline shown)

Fix: Add a regression / trendline overlay with R^2 value; for on-going tracking, use an interactive scatter (tooltip per dist) updated each growing season.

Case - Study : 2

a) **Nature of Dataset** : Single variable tied to districts → geospatial / Categorical.
Rules out line / Scatter (no Continous x-axis)

b) **Chloropleth** : Shows spatial Spread (Villupuram 6.4 vs Coimbatore 2.6) ; good for regional planning.

→ **Weak** : hard to distinguish close values (4.0 / 4.3 / 4.5) , needs map data

Ranked bar : Precise value Comparison ; Clear ranking ; The gaps are immediately quantifiable.

→ **Weak** : no spatial Context , unhelpful for logistics

c) Pattern

→ Highest : Villupuram 6.4 (outlier +1.7 above next).
→ Lowest : Coimabatore 2.6
→ Mid - Cluster : Thanjavur - Dindigul (4.0 - 4.7)

d) Recommendation : Chloropleth map

→ data is inherently district - level
→ enables regional budget / staffing decision via spatial clustering.
→ Villupuram outlier more meaningful shown geographically.

e) Limitation & fix

→ Chloropleth can't precisely convey exact values (4.0 vs 4.3 vs 4.5) look similar in shade).

Fix : Add numeric labels / tooltips on each district or pair chloropleth with a small ranked bar chart as a side panel ; for ongoing tracking , make it interactive.

ANNEXURE : VISUALIZATION

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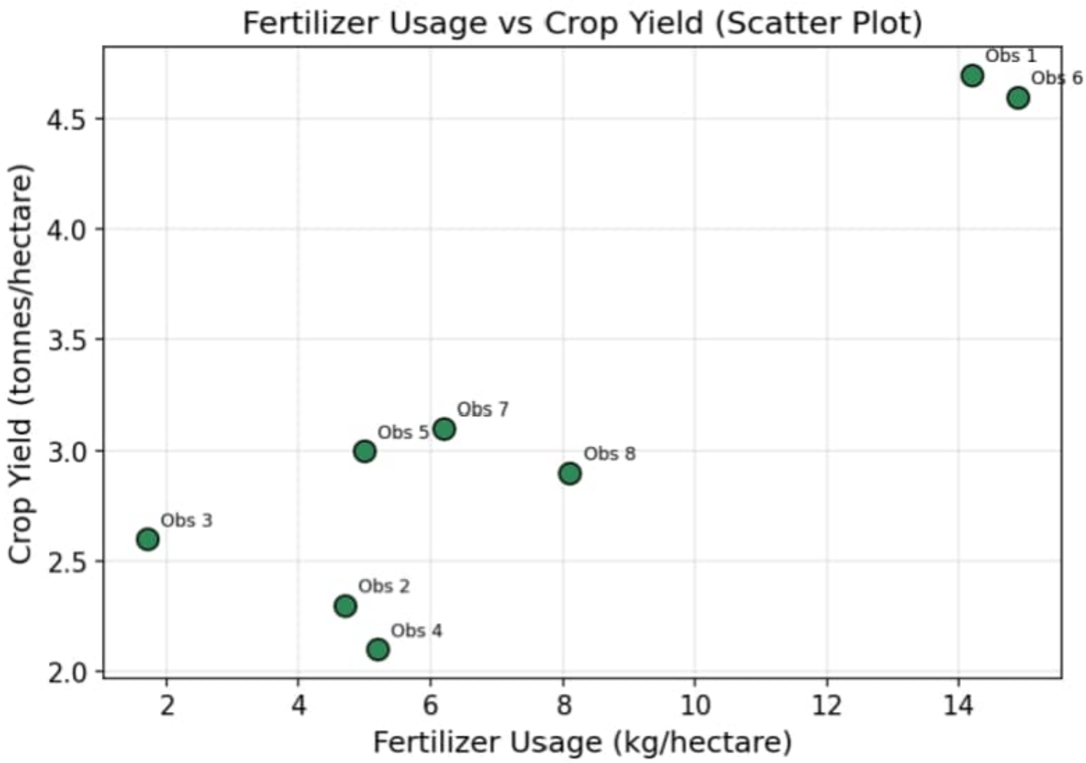
DSA0602 Data Handling and Visualization — AT3 Case Study (BT4)

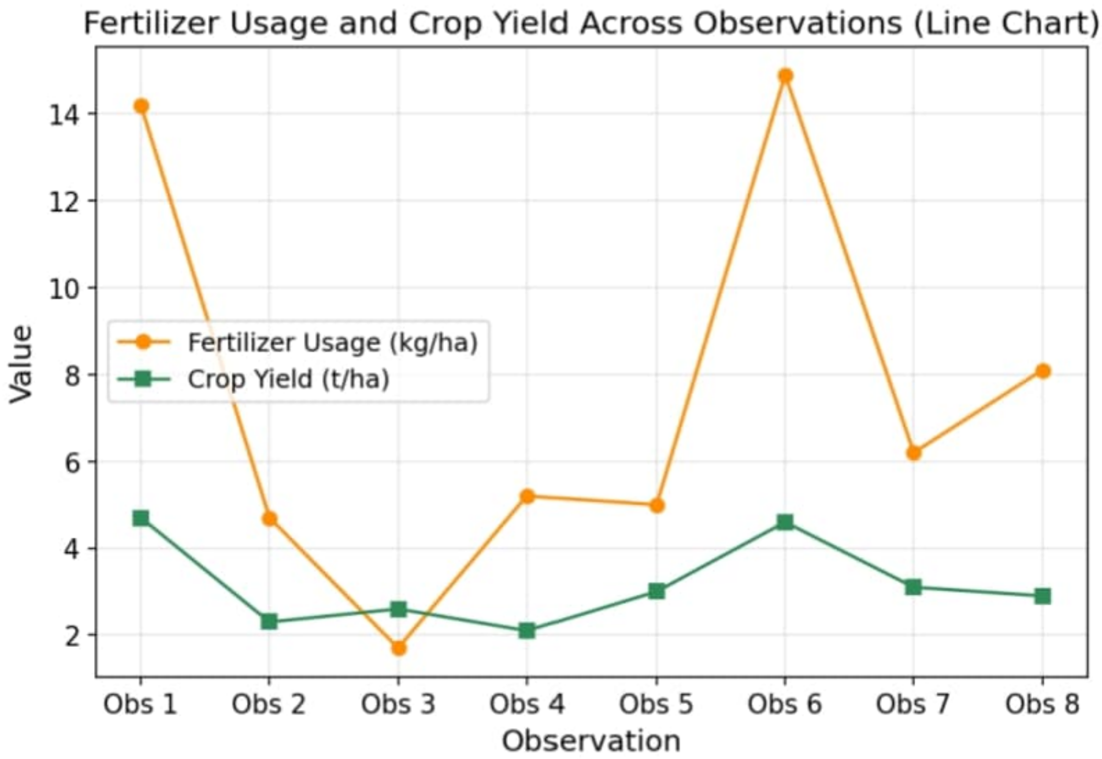
Case Study 1: Fertilizer Usage vs Crop Yield

Topic 1: Directory of Visualizations — x–y Relationships

Scatter Plot



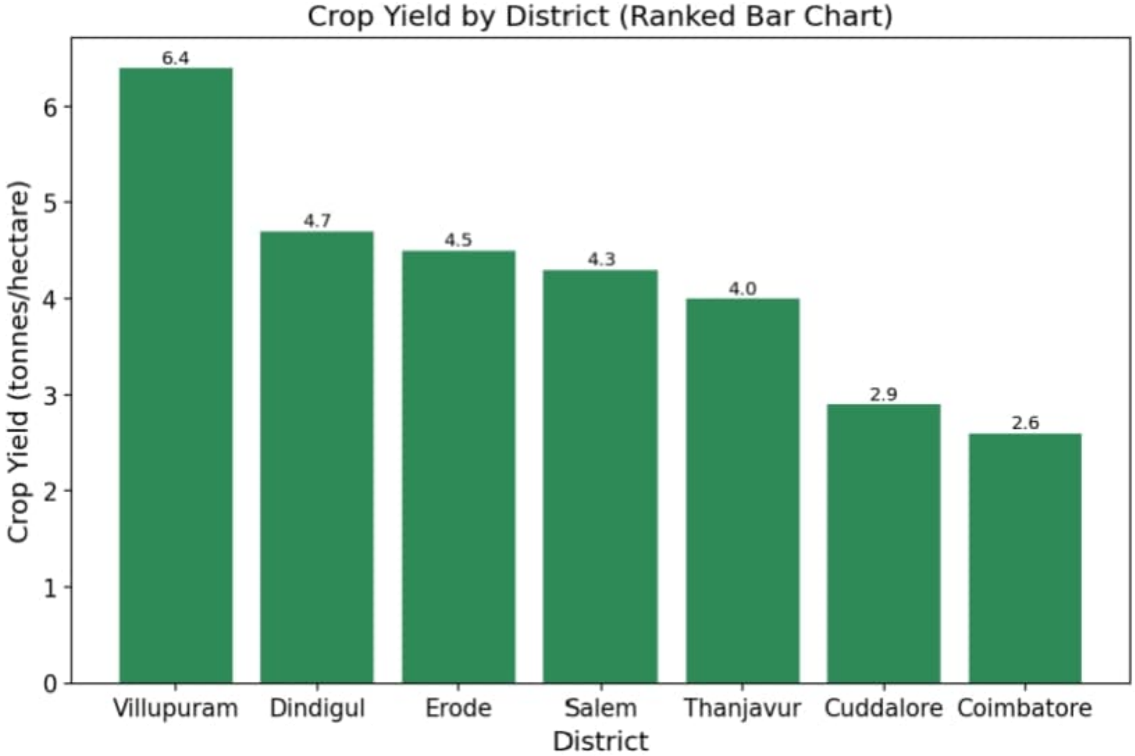
Line Chart


Fertilizer Usage and Crop Yield Across Observations (Line Chart)
Fertilizer Usage (kg/ha)
Crop Yield (t/ha)
Value
Observation
14
12
10
8
6
4
2
Obs 1
Obs 2
Obs 3
Obs 4
Obs 5
Obs 6
Obs 7
Obs 8

Case Study 2: District Crop Yield

Topic 1: Directory of Visualizations — Geospatial Data

Ranked Bar Chart



Geospatial (Map-style) View